

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Game Based Learning Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A Game Based Learning Assessment

Code of Conduct:

I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P Surendra Kumar Reddy

Criterion	Weightage	Marks Obtained
Understanding of the Problem / Game Objective	10%	
Application of AI Search / Problem-Solving Technique	15%	
Successful Completion of the Game / Puzzle	20%	
Efficiency of Solution / Optimal Moves	10%	
Documentation / Evidence of Completion	15%	
Understanding of the Problem / Game Objective	15%	
Originality and Critical Thinking	15%	

Total	100%	
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Signature of the Faculty

Total Marks Awarded

Answer All the Questions

Q. No.	Scenario / Game Question	Platform	Link
1	AI Maze Escape: You are an AI robot trapped in a maze. Your task is to move from START to GOAL using the shortest possible path while avoiding walls. Challenge: Find the shortest path and record the number of steps taken.	Scratch	https://scratch.mit.edu
2	N-Queens Challenge: You are an AI chess strategist. Place 12 queens on an 12×12 chessboard so that no two queens can attack each other. Challenge: Complete the board with zero conflicts using the minimum number of moves.	Online N-Queens Game	https://www.n-queens.com
3	Water Jug Puzzle: You are an AI agent solving a water jug problem. You have two jugs with capacities of 11 litres and 9 litres. Neither jug has measurement markings. Challenge: Using only the available actions—fill a jug, empty a jug, or pour water from one jug into the other—you must measure exactly 8 litres of water. Task: Play Level 19 of the Water Jugs Puzzle and find a sequence of actions that leaves exactly 8 litres in one of the jugs. Try to solve the puzzle using the minimum number of moves.	Tic-Tac-Toe	https://playtictactoe.org

4	Connect Four AI Challenge: You are playing against an AI opponent. Your goal is to connect four of your pieces in a row before the AI does. Challenge: Plan your moves carefully and defeat the AI opponent.	Connect Four	https://www.mathsisfun.com/games/connect4.html
5	8-Puzzle AI Challenge: You are given a scrambled 8-puzzle. Move the tiles one at a time and arrange them in the correct order. Challenge: Solve the puzzle using the minimum possible number of moves.	8-Puzzle Game	https://8puzzle.org/

Structure of the Report – Game-Based Learning

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of the Game-Based Activity	Provide the name of the game or puzzle and the level/challenge attempted.
2	Game Platform / Link	Mention the platform or website used and provide the corresponding game link.
3	Objective of the Activity	Explain the objective of the game and the target that needs to be achieved.
4	Problem Statement	Describe the problem or challenge given in the game. What is the student required to solve?
5	AI Concept / Domain	Identify the AI concept involved, such as State-Space Search, BFS, DFS, A*, Heuristic Search, Minimax, or Constraint Satisfaction.
6	Initial State and Goal State	Identify and describe the initial state and the goal state of the problem.
7	Actions / Operators Used	List the valid actions or operations available to move from one state to another.
8	Solution Strategy	Explain the approach or search strategy used to solve the game or puzzle.
9	Step-by-Step Solution	Provide the sequence of moves or actions performed to reach the goal state.
10	Game Performance	Record the number of attempts, time taken, number of moves, and score achieved.
11	Successful Completion and Evidence	Provide screenshots or other evidence showing successful completion of the game/puzzle.
12	Efficiency and Optimality	Analyse whether the solution was optimal or efficient. Can the puzzle be solved using fewer moves or less time?
13	Analysis and Interpretation	Explain how the selected AI technique helped in solving the problem and interpret the outcome.
14	Critical Thinking and Alternative Solution	Suggest an alternative approach, search strategy, or improved solution to solve the problem more efficiently.
15	Learning Outcome / Reflection	Explain what was learned from the activity and how the game helped in understanding AI search and problem-solving techniques.

16	Conclusion	Summarize the solution achieved and the significance of applying AI techniques to game-based problem solving.
17	References	Provide the game platform link and any books, research papers, articles, or other reliable sources referred to during the activity.

Evaluation Rubrics

Criteria	Percentage (%)	Excellent	Good	Satisfactory	Needs Improvement
Understanding of the Problem / Game Objective	10%	Clearly understands the problem and game objective	Understands the objective with minor errors	Partially understands the problem	Does not understand the objective
Application of AI Search / Problem-Solving Technique	15%	Correctly applies the appropriate AI search or problem-solving technique	Applies the technique with minor errors	Shows limited application of the technique	Unable to apply the appropriate technique
Successful Completion of the Game / Puzzle	20%	Successfully solves/completes the game independently	Completes the game with minimal assistance	Partially completes the challenge	Unable to complete the challenge
Efficiency of Solution / Optimal Moves	10%	Solves using optimal or highly efficient moves	Uses a reasonably efficient sequence of moves	Uses several unnecessary moves	Solution is highly inefficient or incomplete
Documentation / Evidence of Completion	15%	Provides clear steps, score, screenshots, and evidence of completion	Provides most required evidence	Provides incomplete steps or evidence	Provides no proper documentation or evidence

Analysis and Interpretation of the Solution	15%	Clearly analyses the solution and explains the AI technique and outcome	Provides a good analysis with minor gaps	Provides limited analysis or explanation	Unable to analyse or interpret the solution
Originality and Critical Thinking	15%	Demonstrates excellent independent thinking and explores an effective or alternative solution	Demonstrates good independent thinking	Shows limited critical thinking	Shows little or no independent thinking

INTRODUCTION

Artificial Intelligence (AI) uses search algorithms and problem-solving techniques to solve complex tasks efficiently. This assessment provides practical experience with AI-based games that demonstrate concepts such as state-space search, constraint satisfaction, adversarial search, and heuristic search. The activities improve analytical thinking and problem-solving skills through interactive gameplay.

GAME 1 – AI MAZE ESCAPE

1. Title

AI Maze Escape – Shortest Path Search

2. Platform

Scratch

3. Objective

Move the AI character from the start position to the goal using the shortest possible path while avoiding obstacles.

4. Problem Statement

Navigate through the maze and reach the goal using minimum movements.

5. AI Concept

- State Space Search
- Breadth-First Search (BFS)
- Shortest Path Search

6. Initial State

The character starts at the entrance of the maze.

Goal State

Reach the apple (goal).

7. Actions

- Move Up
- Move Down
- Move Left
- Move Right

8. Solution Strategy

Breadth-First Search (BFS) explores the maze level by level and guarantees the shortest path.

9. Step-by-Step Solution

Move through the open paths while avoiding walls until the apple is reached.

10. Game Performance

- Attempts: 1
- Result: Successfully Completed
- Outcome: You Win

11. Evidence

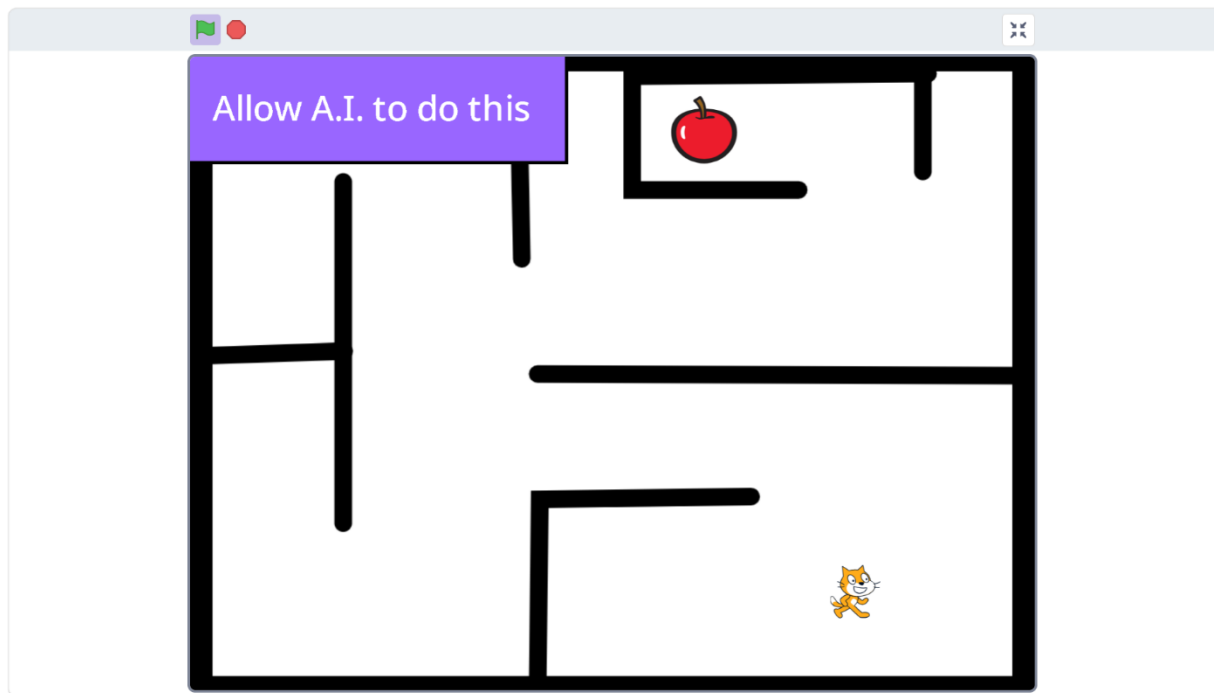


Figure 1: Initial Maze Screenshot

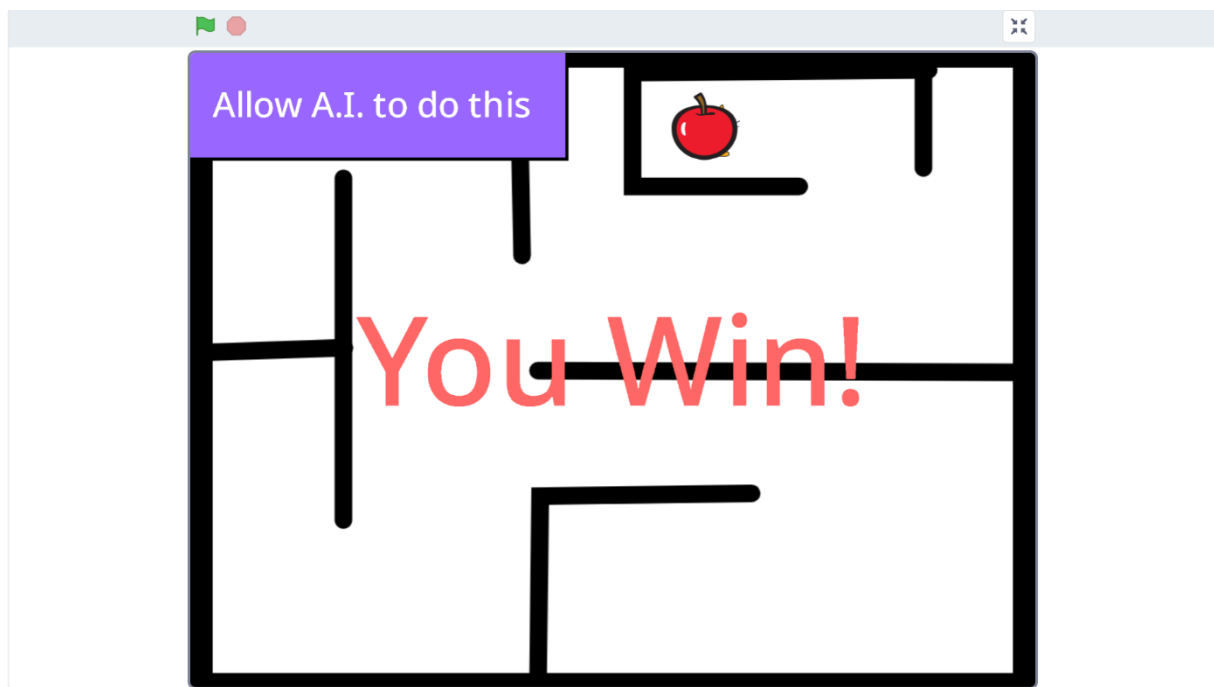


Figure 2: You Win Screenshot

12. Efficiency

The shortest valid path was followed to complete the maze.

13. Analysis

BFS efficiently finds the shortest path between the start and goal.

14. Alternative Solution

A* Search Algorithm.

15. Learning Outcome

Learned how AI finds optimal paths in maze navigation.

16. Conclusion

The maze was successfully solved using shortest-path search techniques.

17. References

Scratch Maze Game

GAME 2 – N-QUEENS CHALLENGE (8×8)

1. Title

N-Queens Challenge (8×8)

2. Platform

Online N-Queens Game

3. Objective

Place eight queens so that no two queens attack each other.

4. Problem Statement

Arrange all queens without sharing the same row, column, or diagonal.

5. AI Concept

- Constraint Satisfaction Problem
- Backtracking
- State Space Search

6. Initial State

Empty 8×8 chessboard.

Goal State

Eight queens placed successfully without conflicts.

7. Actions

- Place Queen
- Remove Queen
- Try another position

8. Solution Strategy

Backtracking places queens one by one and removes incorrect placements until a valid solution is found.

9. Step-by-Step Solution

Place queens row by row while checking conflicts.

10. Game Performance

- Board Size: 8×8
- Queens Placed: 8
- Conflicts: 0
- Result: Successfully Completed

11. Evidence

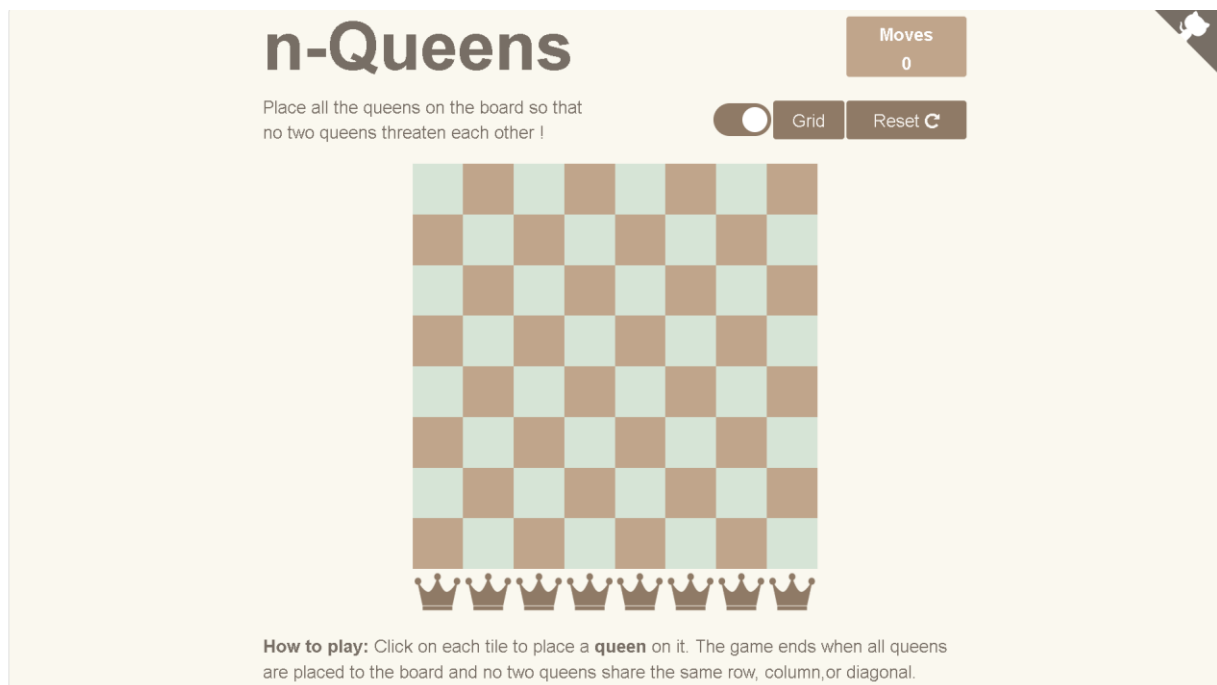


Figure 1: Empty Board

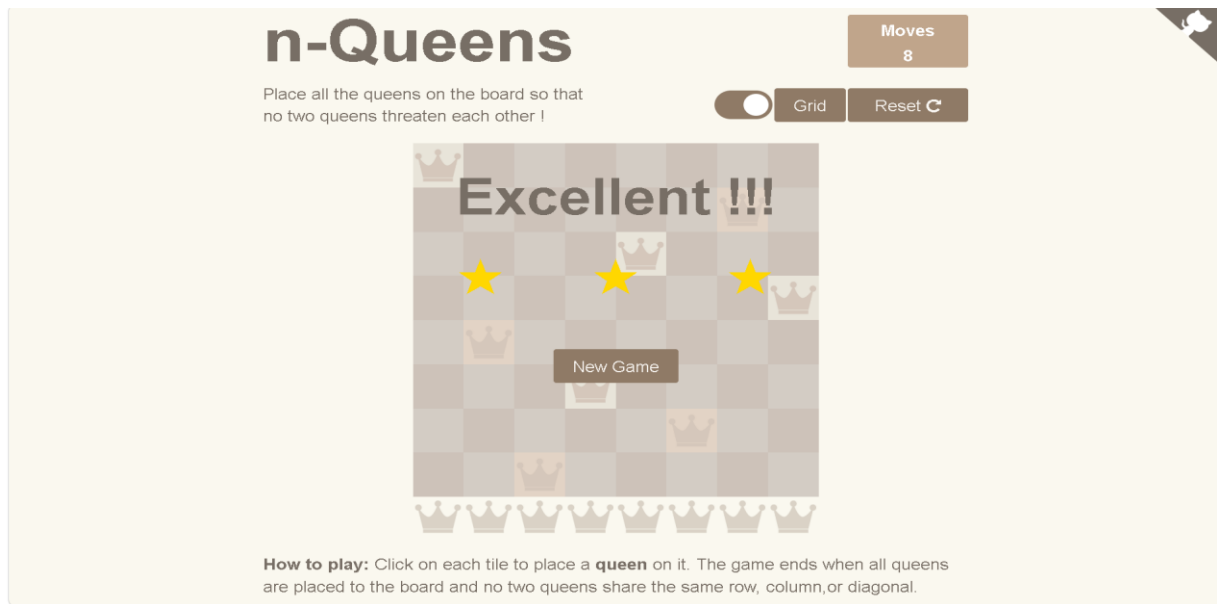


Figure 2: Completed Board

12. Efficiency

The solution satisfies all constraints.

13. Analysis

Backtracking eliminates invalid states efficiently.

14. Alternative Solution

Min-Conflicts Heuristic

15. Learning Outcome

Understood constraint satisfaction and backtracking algorithms.

16. Conclusion

Successfully completed the N-Queens challenge.

17. References

Online N-Queens Game

GAME 3 – ALL (TIC-TAC-TOE)

1. Title

ALL (Tic-Tac-Toe Against AI)

2. Platform

Online Tic-Tac-Toe

3. Objective

Defeat the AI opponent by making three consecutive X symbols.

4. Problem Statement

Play against the computer and make strategic moves to win or draw.

5. AI Concept

- Minimax Algorithm
- Adversarial Search
- Game Playing

6. Initial State

Empty 3×3 board.

Goal State

Three X symbols in a row before the AI.

7. Actions

- Place X
- AI places O

8. Solution Strategy

The AI uses Minimax to determine the best possible move.

9. Step-by-Step Solution

Play strategically by choosing the center or corners and blocking the AI when necessary.

10. Game Performance

- Attempts: 1
- Opponent: AI
- Result: Successfully Completed

11. Evidence

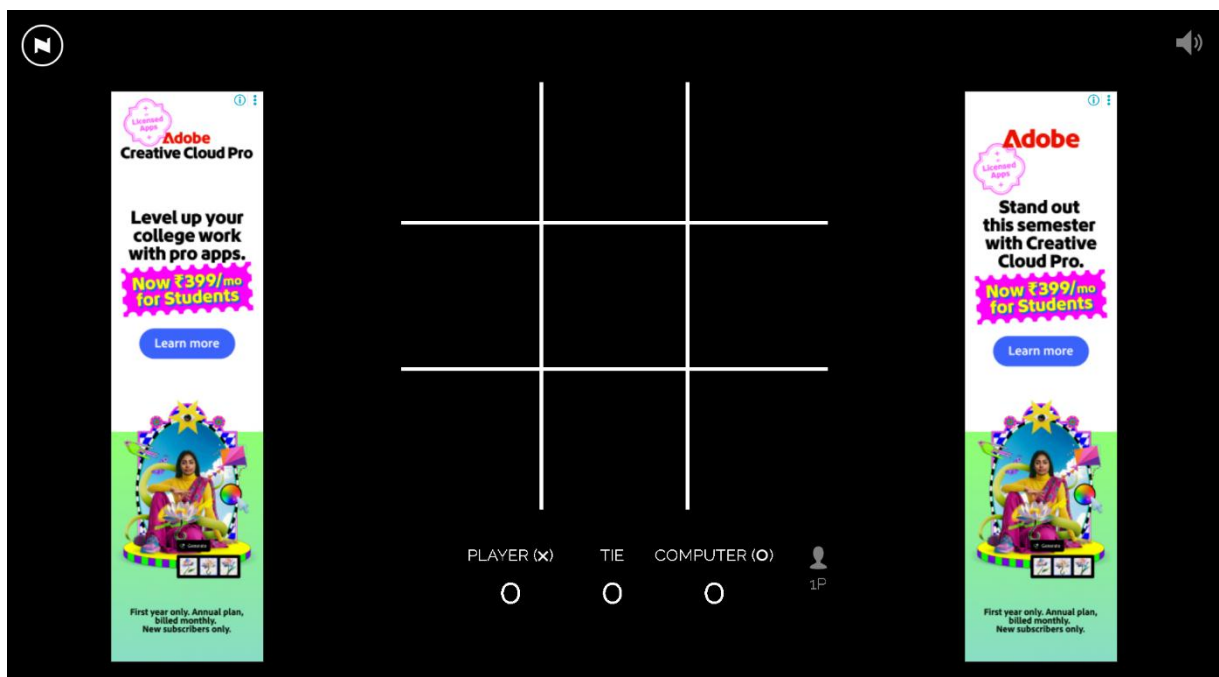


Figure 1: Initial Board

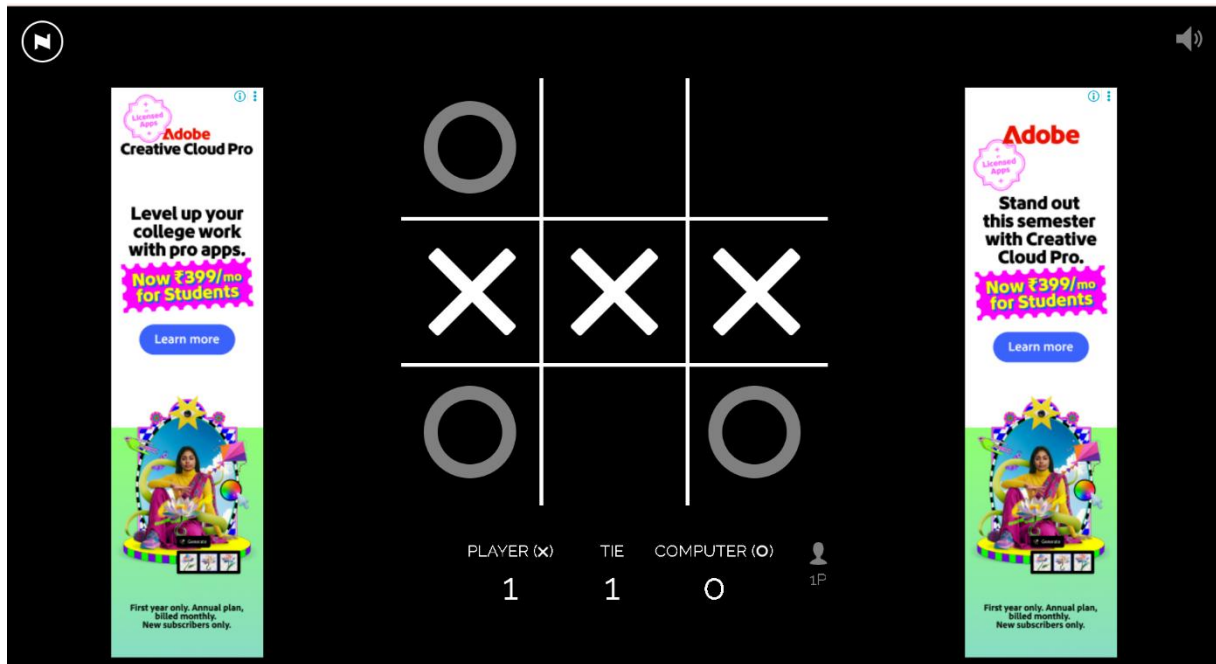


Figure 2: Final Result

12. Efficiency

The AI makes optimal decisions using Minimax.

13. Analysis

The game demonstrates adversarial search between two players.

14. Alternative Solution

Monte Carlo Tree Search (MCTS)

15. Learning Outcome

Learned how AI makes decisions in competitive games.

16. Conclusion

Successfully completed the Tic-Tac-Toe challenge.

17. References

Online Tic-Tac-Toe Game

GAME 4 – CONNECT FOUR AI CHALLENGE

1. Title

Connect Four AI Challenge

2. Platform

Online Connect Four

3. Objective

Connect four pieces before the AI opponent.

4. Problem Statement

Strategically place pieces to form four consecutive discs while blocking the AI.

5. AI Concept

- Minimax
- Alpha-Beta Pruning
- Adversarial Search

6. Initial State

Empty Connect Four board.

Goal State

Four consecutive discs horizontally, vertically, or diagonally.

7. Actions

- Drop Disc
- AI Move

8. Solution Strategy

Use Minimax with Alpha-Beta Pruning to select optimal moves.

9. Step-by-Step Solution

Plan moves while preventing the opponent from connecting four discs.

10. Game Performance

- Attempts: 1
- Result: Successfully Completed

11. Evidence

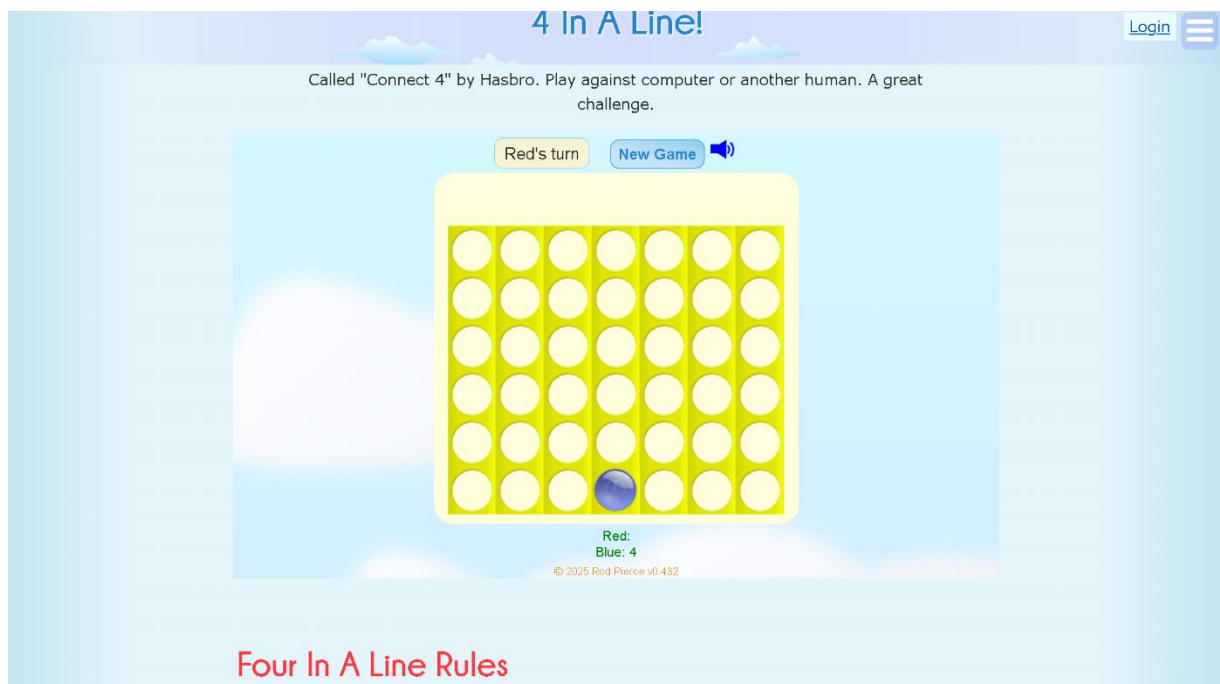


Figure 1: Initial Board

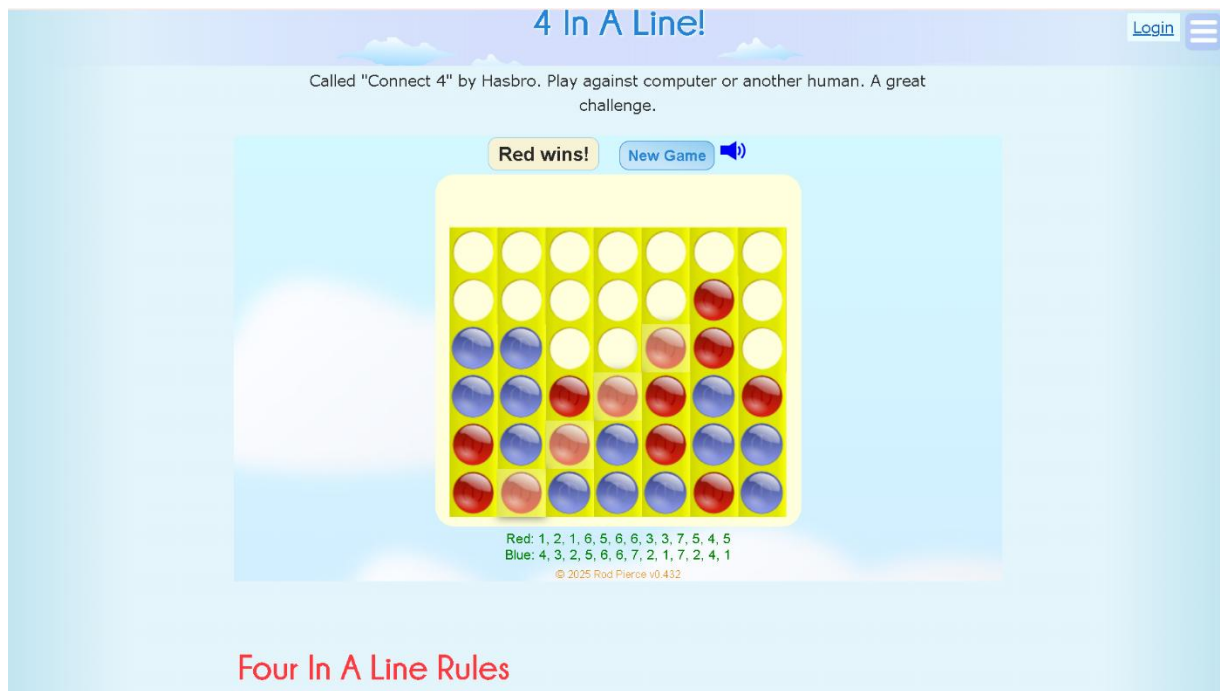


Figure 2: Final Result

12. Efficiency

Strategic play minimizes unnecessary moves.

13. Analysis

The game demonstrates adversarial search and decision-making.

14. Alternative Solution

Monte Carlo Tree Search

15. Learning Outcome

Learned AI strategies in two-player games.

16. Conclusion

Successfully completed the Connect Four challenge.

17. References

Online Connect Four Game

GAME 5 – Sliding Puzzle (5×5)

1. Title

Sliding Puzzle (5×5)

2. Platform

Puzzle Game Hub

3. Objective

Arrange all numbered tiles in ascending order by sliding one tile at a time into the empty space.

4. Problem Statement

The tiles are initially shuffled. The objective is to rearrange them into the correct numerical order using the minimum possible moves.

5. AI Concept

- Heuristic Search
- State Space Search
- A* Search Algorithm

6. Initial State

The 5×5 puzzle board contains 24 numbered tiles placed randomly with one empty space.

7. Goal State

Arrange the tiles in numerical order from **1 to 24**, with the empty space in the final position.

8. Actions / Operators

- Move a tile Up

- Move a tile Down
- Move a tile Left
- Move a tile Right
- Only tiles adjacent to the empty space can be moved.

9. Solution Strategy

The puzzle is solved by moving tiles one at a time into the empty space. AI search algorithms such as **A*** use heuristic functions (for example, Manhattan Distance) to find the shortest sequence of moves efficiently.

10. Game Performance

- Puzzle Size: **5×5**
- Number of Tiles: **24**
- Attempts: **1**
- Result: **Successfully Completed**

11. Evidence

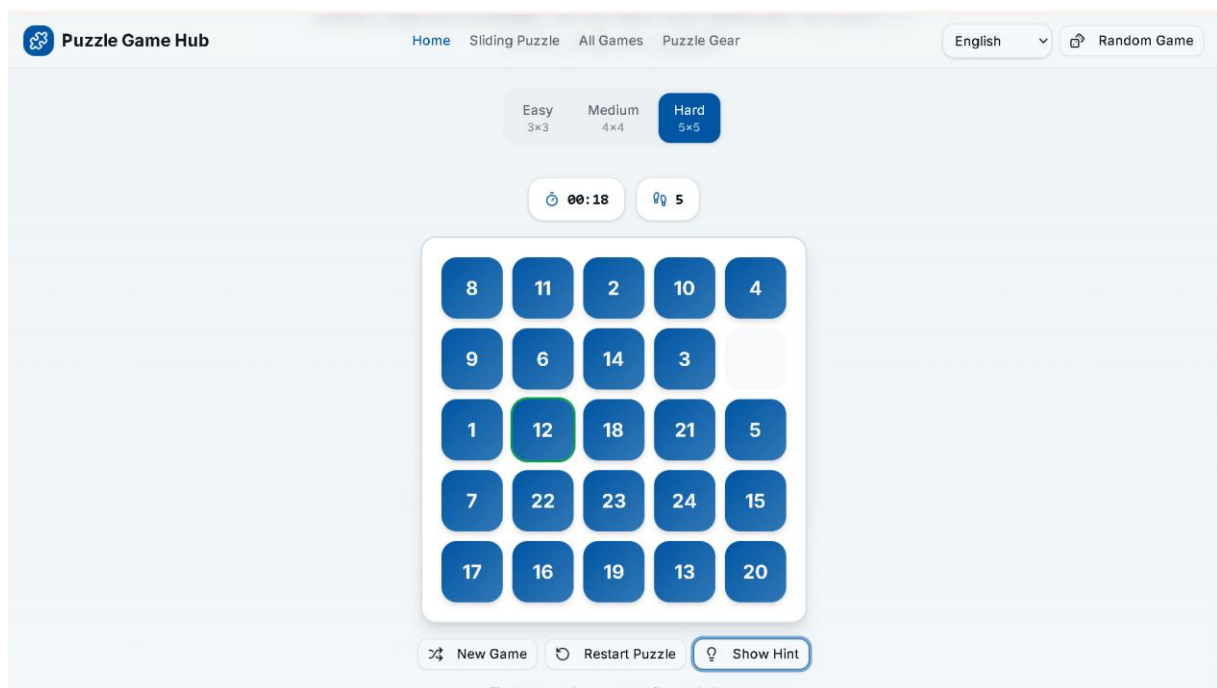


Figure 1: Initial Sliding Puzzle Screenshot

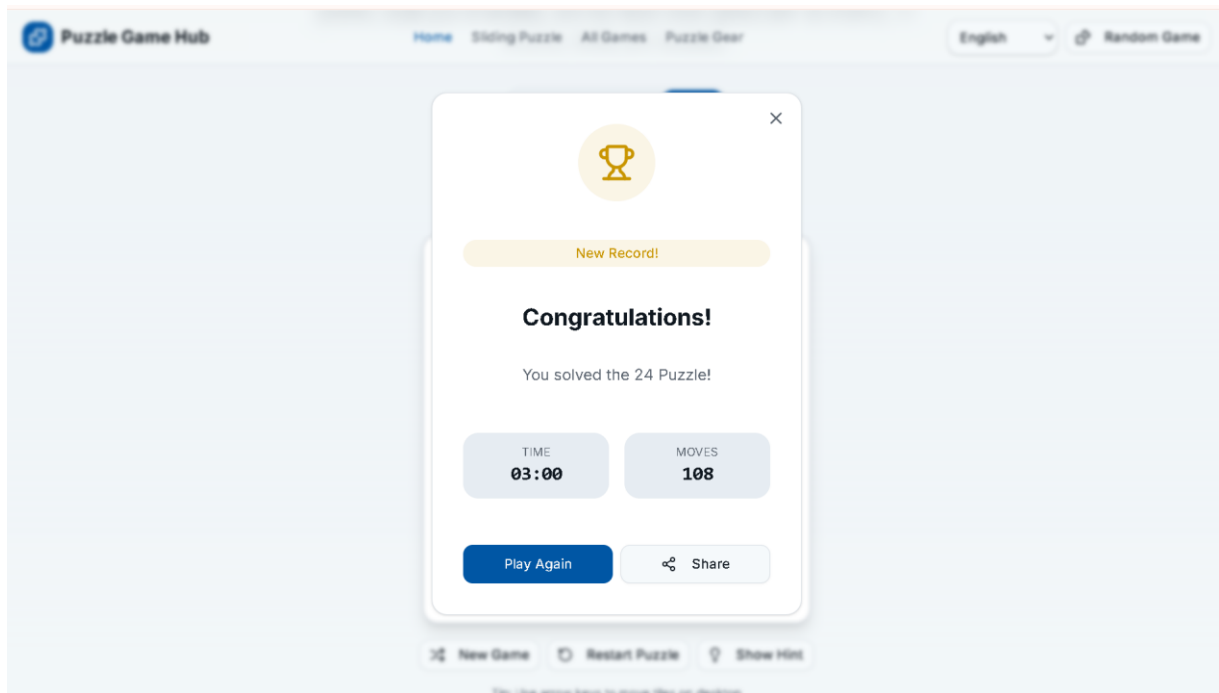


Figure 2: Completed Sliding Puzzle Screenshot

12. Efficiency

The puzzle was solved using logical tile movements while minimizing unnecessary moves.

13. Analysis

The Sliding Puzzle demonstrates heuristic search, where the AI evaluates possible states and selects moves that bring the puzzle closer to the goal configuration.

14. Alternative Solution

- Breadth-First Search (BFS)
- Iterative Deepening A* (IDA*)

15. Learning Outcome

Learned how heuristic search algorithms solve large state-space problems efficiently.

16. Conclusion

The Sliding Puzzle activity demonstrated the practical application of heuristic search techniques for solving complex puzzle problems.

17. References

- Puzzle Game Hub – Sliding Puzzle
- Artificial Intelligence and Expert Systems Lab Manual

OVERALL CONCLUSION

This Game-Based Learning Assessment provided practical knowledge of Artificial Intelligence search techniques. Different AI algorithms such as **Breadth-First Search, Backtracking, Minimax, Alpha-Beta Pruning, and A* Search** were applied to solve various games and puzzles. These activities enhanced problem-solving, logical reasoning, and decision-making skills while demonstrating the real-world applications of AI search algorithms.

REFERENCES

1. Scratch Maze Game
2. Online N-Queens Game
3. Online Tic-Tac-Toe Game
4. Online Connect Four Game
5. 8-Puzzle Game
6. Artificial Intelligence and Expert Systems Lab Manual